41 Nuclear fusion and its containment

Nuclear fusion is the joining together, or 'fusing', of the nuclei of light atoms to make nuclei of heavier atoms. Since the nuclei of the heavier atoms are more stable, energy is released in the process. Under the right conditions, this could be a considerable amount of energy; for instance it provides the energy radiated in vast amounts by the sun and other stars, and the terrifying destructive power of hydrogen bombs.

There are many possible reactions, but one which is the subject of extensive research for peaceful power production is:

$$^{2}_{1}\text{H} + ^{3}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + ^{1}_{0}\text{n} + \text{energy}$$

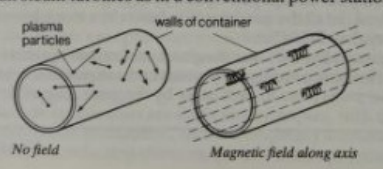
$^{2}_{1}$H is an isotope of hydrogen, called deuterium, and occurs naturally in the hydrogen in sea water, though only to the extent of 1 part in 45 000. $^{3}_{1}$H is another isotope of hydrogen called tritium. This is too rare to be extracted from natural hydrogen, but can be made by bombarding lithium–6 with neutrons:

$$^{6}_{3}\text{Li} + ^{1}_{0}\text{n} \rightarrow ^{3}_{1}\text{H} + ^{4}_{2}\text{He} + \text{energy}$$

The neutrons to bombard the lithium could be provided by the fusion reaction once it has been started, so the basic fuels would be deuterium and lithium. Enough of these are available to give ample energy for millions of years! Another advantage is that none of the final products is radioactive, although tritum is radioactive, with a half-life of 12.4 years.

Unfortunately there are many very difficult engineering problems to be solved before nuclear fusion can be controlled on a big enough scale to provide useful energy.

The main difficulty is that nuclei of any atoms are positively charged and repel each other very strongly, and so have to be travelling at enormous speeds before they collide strongly enough for the short range 'strong nuclear force' to make them fuse. If the nuclei were at an extremely high temperature, say 100 000 000 K, the collisions between them would cause enough nuclei to fuse to provide enough energy to maintain the high temperature *and* give out useful extra energy. This would be in the form of radiation which could be used, indirectly, to boil water to run steam turbines as in a conventional power station.



At this temperature the gas cannot be contained by any known material! Instead it has to be kept from touching the walls of the container by using a magnetic field. The particles of a gas at this temperature consist of electrons and nuclei separated from each other (properly called a plasma), and these charged particles moving at high speed are deflected by the magnetic field as shown in the diagrams.

Data: charge on electron, $e = 1.6 \times 10^{-19}$ C
$\frac{1}{4\pi\varepsilon_o} = 9 \times 10^{9}\ \text{Nm}^2\text{C}^{-2}$
for nucleus at temperature T, K.E. $\approx$ kT
Boltzmann's constant, $k = 1.38 \times 10^{-23}\ \text{JK}^{-1}$
mass of neutron = 1.008986 u (atomic mass unit)
mass of $^{2}_{1}$H = 2.014740 u
mass of $^{3}_{1}$H = 3.017005 u
mass of $^{4}_{2}$He = 4.003873 u
mass of $^{6}_{3}$Li = 6.017034 u
1 u = 1.67×10^{-27} kg = 931 MeV

(a) Why is the equivalent for 1 u expressed in kg *and* MeV? (2)
(b) By consisting the mass before and after the reactions, calculate the energy released in each of the reactions listed. (4)
(c) Calculate the force of electrical repulsion between the nuclei in each reaction if they are 10^{-14} m apart. (Force F between two charges q_1 and q_2, distance r apart, is given by

$$F = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r^2}\Big)$$ (3)

(d) Estimate the speed of a lithium nucleus at the temperature stated. (3)
(e) Describe the path taken by a nucleus in the magnetic field if it is travelling
(i) along the field
(ii) perpendicular to the field
(iii) in some direction between the two. (6)
(f) Calculate the strength of magnetic field needed to keep a lithium nucleus contained within a diameter of 0.2 m. (The force F on a charge q moving at speed v in field B is given by $F = Bqv$, and centripetal acceleration $a = v^2/r$) (6)
(g) If the plasma is prevented from touching the sides of the containment vessel, why will they still get very hot? (2)

Total 30 marks

CIE 9702 Physics Paper 4 Style Question

Nuclear fusion and magnetic confinement

Scientists are investigating controlled nuclear fusion as a long-term energy source. In one proposed reactor, deuterium ($^{2}_{1}$H) and tritium ($^{3}_{1}$H) nuclei fuse to form helium and a neutron:

$$^{2}_{1}\text{H} + ^{3}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + ^{1}_{0}\text{n} + \text{energy}$$

Lithium can also take part in reactions in fusion environments:

$$^{6}_{3}\text{Li} + ^{1}_{0}\text{n} \rightarrow ^{3}_{1}\text{H} + ^{4}_{2}\text{He} + \text{energy}$$

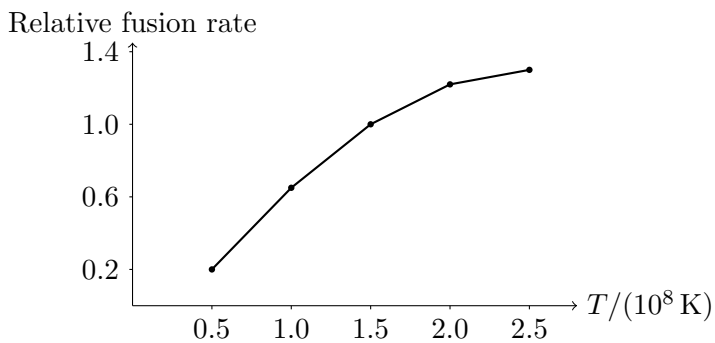
Inside a reactor, the fuel is heated to extremely high temperature so that atoms are ionised and form a plasma. The plasma must be kept away from the vessel walls using magnetic fields. At a plasma temperature $T = 1.5 \times 10^{8}$ K, collisions are frequent and fusion becomes much more likely.

Data

- Charge on electron (magnitude): $e = 1.60 \times 10^{-19}\,\mathrm{C}$
- $\dfrac{1}{4\pi\varepsilon_0} = 9.0 \times 10^{9}\,\mathrm{N\,m^2\,C^{-2}}$
- Boltzmann constant: $k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
- $1\,\mathrm{u} = 1.67 \times 10^{-27}\,\mathrm{kg} = 931\,\mathrm{MeV}/c^2$
- Masses: $m({}^{2}_{1}\mathrm{H}) = 2.01410\,\mathrm{u}$, $m({}^{3}_{1}\mathrm{H}) = 3.01605\,\mathrm{u}$, $m({}^{4}_{2}\mathrm{He}) = 4.00260\,\mathrm{u}$, $m({}^{1}_{0}\mathrm{n}) = 1.00870\,\mathrm{u}$, $m({}^{6}_{3}\mathrm{Li}) = 6.01703\,\mathrm{u}$

For a simplified model of fusion likelihood, the following measured data are obtained in an experiment.

$T/(10^8\,\mathrm{K})$	0.5	1.0	1.5	2.0	2.5
Relative fusion rate (a.u.)	0.20	0.65	1.00	1.22	1.30



Answer all questions. Show your working clearly and quote appropriate units.

1 (a) Explain why extremely high temperature is required before fusion can occur between light nuclei. [2]

(b) State why a plasma cannot be confined by a physical container wall at this temperature and explain how a magnetic field helps. [3]

2 (a) Calculate the energy released, in MeV, in the deuterium–tritium reaction. [3]

(b) Calculate the energy released, in MeV, in the lithium reaction shown. [3]

3 Two nuclei are separated by $r = 1.0 \times 10^{-14}\,\mathrm{m}$.

(a) Calculate the electrostatic repulsive force between two deuterium nuclei at this separation. [2]

(b) A helium nucleus ($+2e$) approaches a lithium nucleus ($+3e$) at the same separation. Determine the new electrostatic force and compare it to your answer in 3(a). [3]

4 Estimate the mean translational kinetic energy of a deuterium nucleus at $1.5 \times 10^8\,\mathrm{K}$ using $E \approx kT$. Hence estimate its speed. [4]

5 In a uniform magnetic field, a charged particle may move:

(i) parallel to the field, (ii) perpendicular to the field, (iii) at an angle to the field.

Describe the path in each case and explain your reasoning briefly. [4]

6 Using your estimated speed from Q4, determine the magnetic flux density required so that a tritium nucleus follows a circular path of radius $0.10\,\mathrm{m}$. Use $F = Bqv$ and centripetal force $F = mv^2/r$. [4]

7 Use the temperature–fusion-rate graph to:

(a) state the temperature range where increasing temperature gives the greatest gain in fusion rate, [2]

(b) suggest why gains become smaller above $2.0 \times 10^8\,\mathrm{K}$. [2]

8 Open-ended question: A reactor designer claims that "raising temperature alone is the best way to increase power output."

Discuss this claim using evidence from the data, your physics knowledge of confinement, and at least **two** other practical factors that affect real reactor performance. [7]

Total: 39 marks

Suggested Answers / Marking Guidance

1(a) Nuclei are positively charged and strongly repel each other electrostatically. Very high temperature gives nuclei high kinetic energy so they can approach close enough for the strong nuclear force to act.

1(b) No solid can withstand direct contact with plasma at $\sim 10^8\,\mathrm{K}$. In plasma, particles are charged, so magnetic force can steer them and keep most of the plasma away from walls.

2(a) $\Delta m = (2.01410 + 3.01605) - (4.00260 + 1.00870) = 0.01885\,\mathrm{u}$. Energy $Q = 0.01885 \times 931 \approx 17.6\,\mathrm{MeV}$.

2(b) $\Delta m = (6.01703 + 1.00870) - (3.01605 + 4.00260) = 0.00708\,\mathrm{u}$. Energy $Q = 0.00708 \times 931 \approx 6.59\,\mathrm{MeV}$.

3(a) $F = \dfrac{1}{4\pi\varepsilon_0}\dfrac{e^2}{r^2} = 9.0 \times 10^9 \times \dfrac{(1.6 \times 10^{-19})^2}{(1.0 \times 10^{-14})^2} \approx 2.3\,\mathrm{N}$.

3(b) Replace $q_1 q_2$ by $(2e)(3e) = 6e^2$, so force is six times Q3(a): $F \approx 13.8\,\mathrm{N}$.

4 $E \approx kT = 1.38 \times 10^{-23} \times 1.5 \times 10^8 = 2.07 \times 10^{-15}\,\mathrm{J}$. Using $E = \frac{1}{2}mv^2$ with $m_D = 2.01410 \times 1.67 \times 10^{-27} \approx 3.36 \times 10^{-27}\,\mathrm{kg}$, $v \approx \sqrt{2E/m} \approx 1.1 \times 10^6\,\mathrm{m\,s^{-1}}$.

5(i) Parallel: straight line (magnetic force zero).
(ii) Perpendicular: circular path (force always perpendicular to velocity).
(iii) At angle: helical path (parallel component unchanged, perpendicular component circular).

6 $Bqv = mv^2/r \Rightarrow B = mv/(qr)$. For tritium, $m_T \approx 3.01605 \times 1.67 \times 10^{-27} = 5.04 \times 10^{-27}\,\mathrm{kg}$, $q = e$, $v \approx 1.1 \times 10^6\,\mathrm{m\,s^{-1}}$, $r = 0.10\,\mathrm{m}$: $B \approx 0.35\,\mathrm{T}$.

7(a) Largest rise is from 0.5 to $1.5 \times 10^8\,\mathrm{K}$ (steepest slope).
7(b) At high T, rate still rises but with diminishing returns; confinement losses, particle escape, and engineering limits become more significant.

8 Indicative points for high-level response: temperature helps but is not sufficient; confinement time and plasma density also determine fusion power; energy losses (radiation, transport), stability and turbulence, magnetic field strength, wall loading, fuel cycle and neutron damage, and net energy efficiency of heating/current drive must be considered. A balanced conclusion should reject the claim as incomplete.

Variant 2 (Higher Difficulty)

Advanced fusion physics and reactor optimisation

A research tokamak is being tested with deuterium–tritium fuel. The engineering team records plasma parameters over several runs while attempting to increase net power output and maintain stable confinement. You are asked to analyse whether the proposed operating point is physically realistic and whether it improves reactor performance.

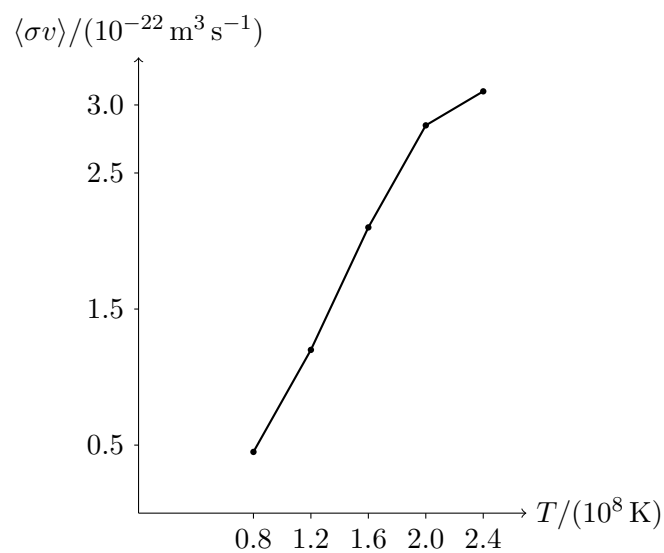
Use the same constants and nuclear masses given earlier in this paper.

Additional data for this variant:

- Vacuum permeability: $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{N\,A^{-2}}$
- Electron-volt conversion: $1\,\mathrm{eV} = 1.60 \times 10^{-19}\,\mathrm{J}$
- Deuterium number density: $n_D = 8.0 \times 10^{19}\,\mathrm{m^{-3}}$
- Tritium number density: $n_T = 7.0 \times 10^{19}\,\mathrm{m^{-3}}$
- Plasma thermal pressure at design point: $p = 2.2 \times 10^{5}\,\mathrm{Pa}$
- Confinement time at design point: $\tau_E = 0.85\,\mathrm{s}$

Experimental reactivity data are approximated by the table below.

$T/(10^8\,\mathrm{K})$	0.8	1.2	1.6	2.0	2.4
$\langle\sigma v\rangle$ $(10^{-22}\,\mathrm{m^3\,s^{-1}})$	0.45	1.20	2.10	2.85	3.10



Answer all questions. Give clear reasoning and appropriate significant figures.

9 (a) Determine the energy released per D–T reaction in joules and in MeV. [3]

(b) If 80% of this energy is carried by the neutron, estimate the neutron energy in MeV. [2]

10 Using the data at $T = 2.0 \times 10^8\,\mathrm{K}$:

(a) calculate the fusion reaction rate density $R = n_D n_T \langle \sigma v \rangle$. [3]

(b) hence estimate the fusion power density $P_f = R E_f$ in $\mathrm{W\,m^{-3}}$. [2]

11 (a) Calculate the magnetic field B needed for beta $\beta = 0.040$, where

$$\beta = \frac{2\mu_0 p}{B^2}.$$

[3]

(b) Explain one operational disadvantage of requiring very large magnetic field strength in a reactor. [2]

12 A simplified Lawson-style criterion is written as

$$nT\tau_E > 3.0 \times 10^{21}\,\mathrm{keV\,s\,m^{-3}}.$$

Using $n = 1.5 \times 10^{20}\,\mathrm{m^{-3}}$, $T = 17\,\mathrm{keV}$ and $\tau_E = 0.85\,\mathrm{s}$, show whether this operating point satisfies the criterion. [4]

13 From the reactivity graph, discuss why simply increasing T beyond $2.0 \times 10^8\,\mathrm{K}$ may yield limited improvement in net output. Your answer should reference both plasma physics and engineering constraints. [4]

14 Open-ended design question (higher-order):

A proposal suggests running at the highest temperature in the table while halving confinement time to reduce disruption risk. Evaluate this proposal quantitatively and qualitatively. Your answer should include: (i) a data-based argument using the graph, (ii) implications for power balance, (iii) one materials issue and one fuel-cycle issue. [8]

Variant 2 Total: 31 marks

Suggested Answers / Marking Guidance (Variant 2)

9(a) From earlier, $E_f \approx 17.6\,\mathrm{MeV}$. In joules: $E_f = 17.6 \times 10^6 \times 1.60 \times 10^{-19} \approx 2.82 \times 10^{-12}\,\mathrm{J}$.

9(b) Neutron carries $0.80 E_f \approx 14.1\,\mathrm{MeV}$.

10(a) At $2.0 \times 10^8\,\mathrm{K}$, $\langle \sigma v \rangle = 2.85 \times 10^{-22}\,\mathrm{m^3\,s^{-1}}$.

$$R = (8.0 \times 10^{19})(7.0 \times 10^{19})(2.85 \times 10^{-22}) \approx 1.60 \times 10^{18}\,\mathrm{m^{-3}\,s^{-1}}.$$

10(b) $P_f = RE_f \approx (1.60 \times 10^{18})(2.82 \times 10^{-12}) \approx 4.5 \times 10^6\,\mathrm{W\,m^{-3}}$.

11(a)

$$B = \sqrt{\frac{2\mu_0 p}{\beta}} = \sqrt{\frac{2(4\pi \times 10^{-7})(2.2 \times 10^5)}{0.040}} \approx 3.7\,\mathrm{T}.$$

11(b) Any valid one, e.g. stronger fields require larger superconducting magnet systems, higher structural stress, and increased cryogenic/engineering complexity.

12

$$nT\tau_E = (1.5 \times 10^{20})(17)(0.85) = 2.17 \times 10^{21}\,\mathrm{keV\,s\,m^{-3}}.$$

Since $2.17 \times 10^{21} < 3.0 \times 10^{21}$, criterion is not met.

13 Expected discussion: graph slope decreases above 2.0×10^8 K (diminishing reactivity gain), while transport/radiative losses and magnet/power-system demands continue to rise; therefore net gain may be modest.

14 Indicative points for high-level response: - From table, increasing T from 2.0 to 2.4 raises reactivity only from 2.85 to 3.10 (about 9%). - Halving τ_E can significantly worsen energy balance and may reduce achievable burn or gain. - Materials issue: higher neutron wall loading, embrittlement, thermal fatigue. - Fuel-cycle issue: tritium breeding ratio, handling losses, processing throughput. - Strong answers weigh trade-offs and conclude that the proposal may improve stability but likely harms overall net performance unless compensated elsewhere.